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Student Number

2015

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT 1**

Extension 1 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time 60 minutes
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A Reference Sheet is provided at the back of this paper
- In Questions 5-7 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 40

Section I - Pages 2 - 3

4 marks

- Attempt Questions 1 – 4
- Allow about 6 minutes for this section

Section II - Pages 4- 7

36 marks

- Attempt Questions 5 – 7
- Allow about 54 minutes for this section

Question	Marks
1 - 5	/4
6	/11
7	/12
8	/13
Total	/40

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 10% of the Higher School Certificate Course Assessment

Section I

4 marks

Attempt Questions 1 – 4

Allow about 6 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 4 (Detach from paper)

1. What is the value of $\int_0^1 \sqrt{1-y^2} dy$?

A. 2π

B. π

C. $\frac{\pi}{2}$

D. $\frac{\pi}{4}$

2. The Cartesian equation of the curves whose parametric equations are $x = \sin t$ and $y = \cos^2 t + 1$ is

A. $y = x^2 - 1$

B. $y = 1 - x^2$

C. $y = 2 - x^2$

D. $y = x^2 - 2$

3. Which group of three numbers could be the roots of the polynomial equation $x^3 + ax^2 - 41x + 42 = 0$?

A. 2, 3, 7

B. 1, -6, 7

C. -1, -2, 21

D. -1, -3, -14

4. When the polynomial $P(x) = x^4 - 8x^3 - 7x^2 + 3$ is divided by $x^2 + x$, the remainder is $ax + 3$.

What is the value of a ?

- A. -14
- B. -11
- C. -2
- D. 5

Section II

36 marks

Attempt Questions 5 – 7

Allow about 54 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 5 – 7, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (11 marks) Use a SEPARATE writing booklet

(a)

The cubic equation $x^3 + 5x^2 + cx + d = 0$ has three roots; -3 , 7 and α .

(i) Use the sum of roots to find α . **1**

(ii) Find the values of “c” and “d”. **1**

(b)

Let $P(x) = x^3 + 3x - 7$

(i) Show that the equation $P(x) = 0$ has a root between 1 and 2. **2**

(ii) Use 2 applications of Newton’s method with initial approximation $x_1 = 1$ to approximate this root. Give your answer correct to two decimal places. **2**

(c) The polynomial equation $P(x) = 0$ has a double root at $x = a$.

(i) By writing $P(x) = (x - a)^2 Q(x)$, where $Q(x)$ is a polynomial, show that $P'(a) = 0$. **2**

(iii) Hence or otherwise, find the values of a and b if $x = 1$ is a double root of $x^4 - ax^3 + bx^2 + 5x - 1 = 0$ **3**

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet

- (a) A circle is given by the equation $x^2 + y^2 + 4x - 2y - 20 = 0$ **2**
Find the equation of the locus of the mid points of all the chords of length 8 units.
- (b) The parametric equations of a curve are $x = t^2 + \frac{1}{t^2}$ and $y = t + \frac{1}{t}$. **1**
Find the Cartesian equation of the curve.
- (c) Find a pair of parametric equations for the parabola $x^2 = 12(y - 3)$ **2**
- (d) The parametric equations of a parabola are $x = 4t$ and $y = 2t^2$, where t is a parameter.
- (i) $P(4p, 2p^2)$ is a point on the above parabola. Find the equation of the tangent to the parabola at P . **2**
- (ii) Q is the foot of the perpendicular drawn from the vertex of the parabola to the tangent at P .
- (α) Find the coordinates of Q . **2**
- (β) Show that the locus of Q is given by $y(y^2 + x^2) + 2x^2 = 0$ **3**

End of Question 6

Question 7 (13 marks) Use a SEPARATE writing booklet

(a) Find: $\int (x^3 - \frac{1}{x^2})^2 dx$ 2

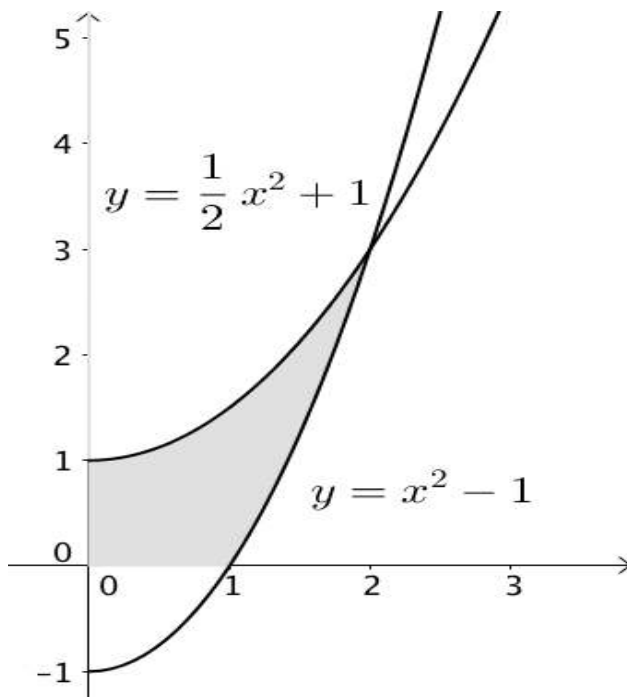
(b) Evaluate: $\int_0^2 \frac{dx}{(3x+2)^2}$ 3

(c) The following functions $f(x) = 3x^2$ and $g(x) = 7x - x^2$ intersect at the origin and the point A.

(i) Find the coordinates of A where the two curves meet. 2

(ii) Calculate the area enclosed by the two curves. 3

(d) A wooden bowl is made by rotating the area between the curves $y = \frac{1}{2}x^2 + 1$ and $y = x^2 - 1$, in the first quadrant about the y-axis.



If the shaded area represents the cross section of the bowl, calculate the volume of wood in the bowl.

3

2015 Task 1 (HSC Ext 1)

- ① Semi circle radius 1

$$\int_0^1 \sqrt{1-y^2} dy = \frac{1}{2} \pi (1)^2 = \pi/2$$

(C)

- ② $x = \sin t$, $y = \cos^2 t + 1$

$$= (1 - \sin^2 t) + 1 \\ = 2 - x^2$$

(C)

- ③ Could be A or C as needs to be factors of 42 and be positive when multiplied by since there is a term $-41x \rightarrow$ must have negative

~~(A)~~

(B)

- ④ By long division

$$\begin{array}{r} x^2 - 9x + 2 \\ x^2 + x \overline{) x^4 - 8x^3 - 7x^2 + 3} \\ \underline{x^4 + x^3} \\ -9x^3 - 7x^2 \\ \underline{-9x^3 - 9x^2 + 0x} \\ 2x^2 + 2x \\ \underline{-2x + 3} \end{array}$$

(C)

- ⑤ a) i) $x^3 + 5x^2 + cx + d = 0$
roots $-3, 7, 2$

$$\therefore -3 + 7 + 2 = -5$$

$$\therefore c = -9$$

✓ (1)

ii) $c = (-3)(7) + (-3)(-9) + (7)(-9)$

(product of roots two at a time)

$$= -57$$

$$d = -3 \times 7 \times -9 = 189$$

✓ (1)

5 b) i) $P(x) = x^3 + 3x - 7$
 $P'(x) = 3x^2 + 3$

(2)

$P(1) = 1 + 3 - 7 = -3$ ✓ (1)

$P(2) = 8 + 6 - 7 = 7$

✓ (1)

Since $P(1) < 0$ and $P(2) > 0$ and $P(x)$ is continuous
 $1 \leq x \leq 2 \rightarrow$ must be a root between $x=1$ and $x=2$

ii) Let $x_1 = 1$

$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$= 1 - \frac{(-3)}{6}$

$= 1\frac{1}{2}$

✓ (1)

$\therefore x_3 = \frac{3}{2} - \frac{\frac{7}{8}}{\frac{39}{4}}$

$= \frac{55}{39}$

✓ (1)

$= 1.41$

c) i) $P(x) = (x-a)^2 Q(x)$ and $P(a) = 0$

$\therefore P'(x) = 2(x-a)Q(x) + (x-a)^2 Q'(x)$ ✓ (1) by the product rule

$= [x-a] (2Q(x) + (x-a) Q'(x))$ ✓ (1)

$\therefore P'(a) = 0 \rightarrow$

ii) $P(x) = x^4 - ax^3 + bx^2 + 5x - 1$
 $P'(x) = 4x^3 - 3ax^2 + 2bx + 5$

✓ (1)

Now $P(1) = 1 - a + b + 5 - 1 = -a + b + 5 = 0$ } (1)

$P'(1) = 4 - 3a + 2b + 5 = 0$ } (2)

✓ (1)

from (1) $b = a - 5$ sub into (2)

$9 - 3a + 2(a - 5) = 0$

$\therefore -a = 1$

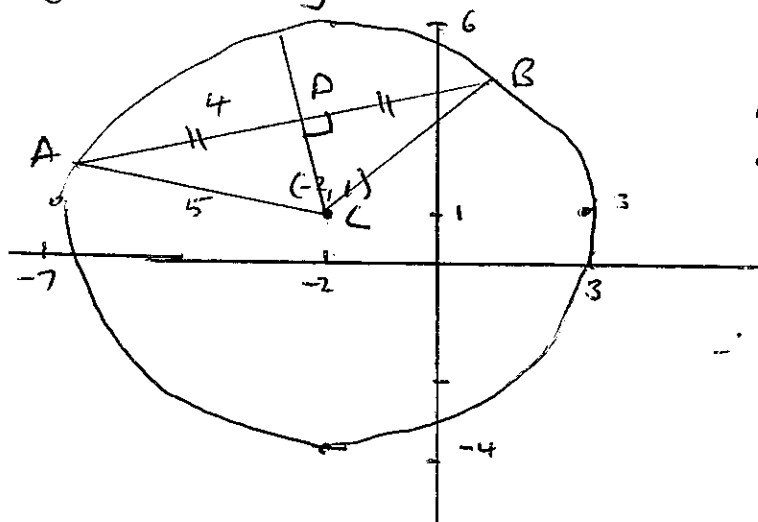
$a = -1$

$\rightarrow b = -6$

✓ (1)

⑥ a) $x^2 + 4x + 4 + y^2 - 2y + 1 = 20 + 5$
 $\therefore (x+2)^2 + (y-1)^2 = 25$

③



let $AB = 8$
 now $AC = 5$
 and CD bisects AB
 at right angles
 (Circle Geo!)

$\therefore CD = 3$ by $\checkmark$ ①
 Pythagoras

$\therefore$ locus of midpoint will
 be a circle centre $(-2, 1)$ radius 3

$\therefore (x+2)^2 + (y-1)^2 = 9$

$\checkmark$ ①

b) $x = t^2 + \frac{1}{t^2}$

$y = t + \frac{1}{t}$

$\therefore y^2 = \left(t + \frac{1}{t}\right)^2$
 $= t^2 + \frac{1}{t^2} + \frac{2t}{2}$
 $= t^2 + \frac{1}{t^2} + 2$

$\checkmark$ ①

$\therefore x = y^2 - 2$

c) $x = 6p, y = 3(p^2 + 1)$

$\checkmark$ ①

show $p = \frac{x}{6} \Rightarrow y = 3\left(\left(\frac{x}{6}\right)^2 + 1\right)$
 $y = \frac{3x^2}{36} + 3$

$\checkmark$ ① for verifying.

$\therefore y - 3 = \frac{x^2}{12}$
 $\therefore 12(y - 3) = x^2$

⑥ a) i) $x = 4t$, $y = 2t^2$

④

$$\rightarrow y = 2 \left(\frac{x}{4} \right)^2$$

$$= \frac{2x^2}{16}$$

$$= \frac{x^2}{8}$$

$\rightarrow$ parabola vertex $(0, 0)$

✓ ①

$$\therefore \frac{dy}{dx} = \frac{x}{4} \quad \text{at } P(4p, 2p^2)$$

$$\frac{dy}{dx} = p$$

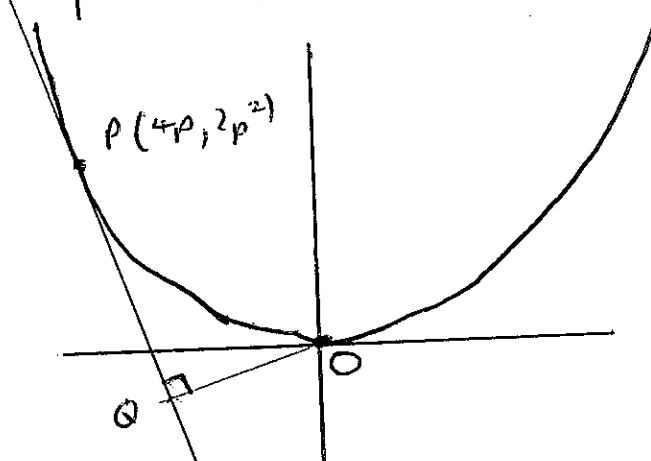
$\therefore$ Eqⁿ of tangent

$$y - 2p^2 = p(x - 4p)$$

✓ ①

$$\therefore y = px - 2p^2$$

ii) a)



line OQ must have gradient $-\frac{1}{p}$ and passes through Q

$$\therefore y = -\frac{1}{p}x$$

find Q by solving simultaneously

$$-\frac{1}{p}x = px - 2p^2$$

$$-x = p^2x - 2p^3$$

$$\therefore 2p^3 = p^2x + x$$

$$= x(p^2 + 1)$$

$$\therefore x = \frac{2p^3}{p^2 + 1}$$

$$\therefore y = \frac{-2p^2}{p^2 + 1}$$

$$13) \quad x = \frac{2p^3}{p^2+1}$$

$$y = \frac{-2p^2}{p^2+1}$$

5

Notice

$$x = -py$$

$$\therefore p = \frac{-x}{y} \rightarrow \text{substitute}$$

✓ 1

$$x = \frac{2 \left(\frac{-x}{y} \right)^3}{\left(\frac{-x}{y} \right)^2 + 1}$$

$$= \frac{\frac{-2x^3}{y^3}}{\frac{x^2}{y^2} + 1}$$

✓ 1

$$x = \frac{\frac{-2x^3}{y^3}}{\frac{x^2+y^2}{y^2}}$$

$$\therefore x = \frac{-2x^3}{y(x^2+y^2)}$$

$$\therefore xy(x^2+y^2) = -2x^3$$

$$\therefore y(x^2+y^2) = -2x^2$$

$$\therefore y(y^2+x^2) + 2x^2 = 0$$

since $x \neq 0$ ✓ 1

$$\textcircled{7} \text{ a) } \int \left(x^3 - \frac{1}{x^2} \right)^2 dx$$

⑥

$$= \int \left(x^6 - \frac{2x^3}{x^2} + x^{-4} \right) dx \quad \checkmark \textcircled{1}$$

$$= \int (x^6 - 2x + x^{-4}) dx$$

$$= \frac{x^7}{7} - \frac{x^2}{2} - \frac{x^{-3}}{3} + c$$

✓①

$$\begin{aligned} \text{b) } \int_0^2 \frac{dx}{(3x+2)^2} &= \left[-\frac{1}{3} (3x+2)^{-1} \right]_0^2 \\ &= \left[-\frac{1}{9x+6} \right]_0^2 \\ &= \left[-\frac{1}{24} \right] - \left[-\frac{1}{6} \right] \\ &= \frac{1}{8} \end{aligned}$$

$$\text{c) } f(x) = 3x^2$$

$$\begin{aligned} g(x) &= 7x - x^2 \\ &= x(7-x) \end{aligned}$$

∴ Intersect when

$$3x^2 = 7x - x^2$$

$$\therefore 4x^2 - 7x = 0$$

$$x(4x-7) = 0$$

intersect

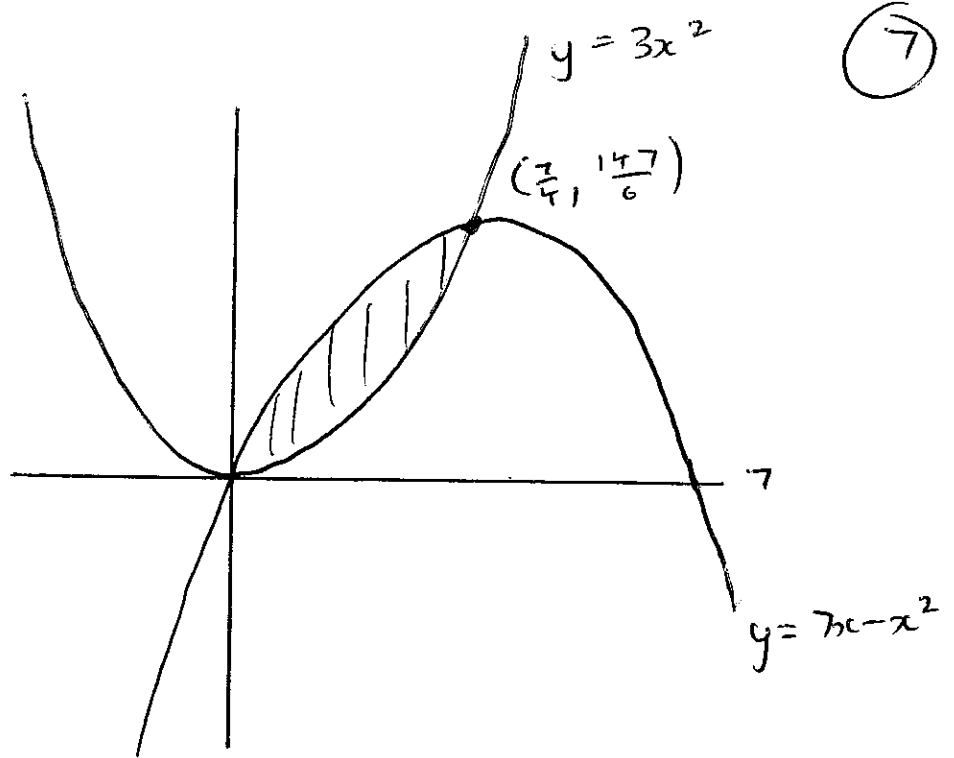
$$x=0, \quad x=\frac{7}{4}$$

$$\text{when } x = \frac{7}{4}, \quad y = 3 \left(\frac{49}{16} \right) = \frac{147}{16}$$

Curves meet at

$$\left(\frac{7}{4}, \frac{147}{16} \right)$$

7c) continued



$$A = \int_0^{7/4} (7x - x^2) - (3x^2) dx \quad \checkmark \textcircled{1}$$

$$= \int_0^{7/4} (7x + 2x^2) dx$$

$$= \left[\frac{7}{2}x^2 + \frac{2}{3}x^3 \right]_0^{7/4} \quad \checkmark \textcircled{1}$$

$$= \left[\frac{7}{2} \left(\frac{49}{4} \right) + \frac{2}{3} \left(\frac{343}{8} \right) \right] - 0$$

$$= \left[\frac{343}{8} + \frac{686}{24} \right]$$

$$= \frac{1715}{24} \text{ units}^2 \quad \checkmark \textcircled{1}$$

$$7d) \quad y = \frac{1}{2}x^2 + 1 \quad , \quad y = x^2 - 1$$

⑧

meet when

$$x^2 - 1 = \frac{1}{2}x^2 + 1$$

$$\therefore \frac{1}{2}x^2 = 2$$

$$x^2 = 4$$

$$x = \pm 2$$

→ we only want $x = 2$

$$\therefore y = 3$$

✓ ①

$$\text{Volume} = \pi \int_0^3 x^2 \cdot dy$$

where x^2 is the radius of the disc

Now from the graph the region we need has radius

$$y = x^2 - 1$$

$$\therefore x^2 = y + 1$$

$$y = \frac{1}{2}x^2 + 1$$

$$y - 1 = \frac{1}{2}x^2$$

$$x^2 = 2y - 2$$

$\therefore$ radius of disc

$$(y+1) - (2y-2)$$

OR

$$V = \pi \int_0^3 (y+1) \cdot dy - \pi \int_0^3 (2y-2) dy$$

✓ ①

$$= \pi \int_0^3 (-y + 3) \cdot dy$$

$$= \pi \left[-\frac{y^2}{2} + 3y \right]_0^3$$

$$= \pi \left[-\frac{9}{2} + 9 \right]$$

$$= \frac{9\pi}{2} \text{ units}^3$$

✓ ①